

Udeshya Kumar

A learner with a strong interest in full-stack development, software engineering, and system design

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SUMMARY

Full-stack developer and data engineer with hands-on experience building scalable data pipelines, backend systems, and web applications. Recently graduated with a B.Tech in Computer and Communication Engineering, with a minor specialization in Full-Stack Development and Big Data. Skilled across the modern JavaScript/Python ecosystem, cloud data platforms (Snowflake, AWS, Airflow), and system design, with a track record of shipping production-grade features and personal projects end to end.

WORK EXPERIENCE

Bain Capabilities Network

Jan 2026 – Jul 2026

Data Analyst Intern

- Designed a platform-agnostic pipeline controlled through Airflow and capable of executing workloads across Snowflake and Starburst, helping reduce compute costs.
- Built and maintained Airflow DAGs to manage and schedule data pipelines from AWS S3.
- Developed Snowflake stored procedures for automated data ingestion using SQL, JavaScript, and Python.
- Imported 10+ Snowflake-dependent stored procedures into platform-agnostic Python modules.
- Created Apache Iceberg tables to enable unified data access across both Snowflake and Starburst platforms.

EmergX AI

June 2025 – July 2025

Full Stack Intern

- Created new pages and implemented dynamic routes in Next.js.
- Enhanced frontend data fetching and API interactions in React by implementing custom hooks and optimized state management, improving functionality and responsiveness.
- Designed and implemented API gateway and middleware layer such as JWT-based authentication for secure login and routes to handle user data submission and retrieval, connecting frontend forms to MongoDB for real-time CRUD operations.
- Used Git and GitHub for version control across the codebase, tracking changes through commits, reviewing diffs before merging, and collaborating with the team via branches and pull requests.
- Improved application responsiveness and reduced loading times by 30% identifying and fixing memory leaks.

PROJECTS

Ultimate Tic-Tac-Toe

- Full-stack multiplayer game of nine tic-tac-toe boards in a 3×3 grid, with local and online modes.
- Server-validated moves prevent cheating; shareable room links enable quick multiplayer invites.
- Built with React, TypeScript, and Vite on the frontend, and Node.js, Express, and Socket.IO on the backend.

Food Scanner

- Mobile app to scan food barcodes and fetch nutrition, ingredients, and brand data via the Open Food Facts API.
- Assigns an A–E health grade based on nutritional value and calorie count.
- Built with Dart and Flutter, with SQLite for local storage.

Life-Expectancy Estimator

- Survival-analysis app combining SSA mortality tables with a Bayesian Weibull hazard model (PyMC), validated against a Cox model (lifelines), trained on NHANES health data.
- Predicts individualized survival probability and expected lifespan from age, gender, BMI, blood pressure, smoking, and alcohol use.
- Built the data pipeline in Python/pandas with an interactive Streamlit + Plotly app for posterior survival curves.

SKILLS

Programming Languages: JavaScript&TypeScript, C/C++ , Python, SQL, Java

Frameworks, Tools & Platforms: Git, Bash, Node.js ,React.js, Next.js, Express.js, FastAPI, Docker, Codex, Cursor, AWS

Concepts: Data Structures and Algorithms, Object-Oriented Programming, System Design, Agile Methodology, Relational and Non-Relational Databases, Encryption & Decryption, Operations research

Languages: English, Hindi.

EDUCATION

Manipal Institute of Technology, Manipal

2022 – 2026

B.Tech in Computer and Communication Engineering — CGPA: 7.19

- Minor specialization in Full-Stack Development and Big Data